

DEEPESH SINGH

AI Engineer | AI & Deep Learning Scientist | Agentic AI, LLMs, RAG, Generative AI, Computer Vision

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PROFESSIONAL SUMMARY

AI & Deep Learning Scientist with **4+ years** of experience building production AI products that connect research, engineering, and user-facing delivery. Leads a **4-person** AI engineering and research team and owns the full product lifecycle, from planning and system design to development, testing, deployment, and API integration. Shipped an **audio-visual dubbing studio**, a **multi-tenant enterprise RAG assistant**, and a **multi-agent workflow automation** platform, with Flask/FastAPI backends for UI compatibility. Strong background in **Visual Transformer (ViT)** and **diffusion**-based video pipelines, agentic AI (LangGraph, MCP), RAG, LLM evaluation and observability, cloud infrastructure, and CI/CD.

PROFESSIONAL EXPERIENCE (4+ years in AI/ML)

AI & Deep Learning Scientist — Cynapto Technologies Pvt. Ltd., Mumbai

Aug 2022 – Present

- **Leadership** — led a **4-person** AI engineering and research team across **3 production AI products**: sprint planning, architecture and code reviews, and mentoring.
- **Agentification (in rollout)** — converting the monolithic dubbing platform to microservices with **12+ agents and sub-agents** in LangGraph, using checkpointing for resumable long-running workflows to remove manual intervention from end-to-end dubbing delivery.
- **Agent Platform** — built the multi-agent workflow automation product on LangGraph and Langflow: drag-and-drop pipelines, event triggers, tool/function calling, MCP integrations, human-in-the-loop checkpoints, and LiteLLM routing with fallback chains; **4 production deployments**, 3–6 sub-agents each, running 24x7 with durable state, retries, and idempotent recovery.
- **Enterprise RAG** — designed a multi-tenant RAG Assistant for natural-language queries over organization-specific knowledge bases (LangChain, OpenAI SDK, FAISS/Pinecone/Qdrant, FastAPI, Google and Airtable connectors) with per-org vector namespaces, RBAC, and rate limiting, and query routing across Text, Query, Graph, and Self-RAG strategies; evaluated with Ragas on a **200-question golden set** of internal policy and product documents — **0.88 faithfulness, 0.82 answer correctness** — at **~4s p95 latency**.
- **Evaluation & Observability** — built automated LLM evaluation (LLM-as-judge, golden sets, verdict gating) with Ragas retrieval scoring and agent-run tracing for token, cost, and latency; used Langfuse for prompt registry and agent audits.
- **Dubbing Studio · Lip-Sync Model** — built a production diffusion-based lip-sync model as a three-stage architecture: lip-sync generation, Reference VAE identity preservation, and 4x GAN super-resolution (256px→1024px) with temporal smoothing for broadcast-quality output — the identity-preservation stages cutting **per-speaker training time 86%**; trained on **500+ hours** across **4xH100, 4xA100, and A40** clusters with PyTorch DDP, improving Whisper log-mel spectrogram features over DeepSpeech.
- **Dubbing Studio · Translation Engine** — built the **50+ language** engine driving the studio: context flow across scenes, speaker profiling, and scene interpretation for colloquial, semantically accurate and phonetically synchronized output, with a multi-step judgement-based feedback loop driving automated retries — cutting manual edits **~70%**.
- **Computer Vision** — movie-grade face detection, recognition, and tracking (S3FD, 2DFAN/3DFAN, re-ID fallbacks) and real-time manufacturing activity analysis combining body/hand pose, object detection, and temporal action recognition against defined SOPs.
- **Platform Engineering** — optimized SQL and inference APIs across **90+ endpoints in 3 services** for **3x faster queries**; owned Linux/Nginx deployments, CDN migration, and GitLab CI/CD.

Backend Django Developer — Skyrath Pvt. Ltd., Delhi

Nov 2021 – Jul 2022

- Built a grocery delivery platform's REST API backend from scratch (JWT auth, MySQL/MongoDB, Google Maps, cron jobs, admin dashboard); deployed infrastructure using Docker on AWS with Nginx.

Research Intern — Omdena (Google-partnered open-source research)

Sep 2021 – Nov 2021

- Collaborated with **29 selected researchers** analyzing sensitive government datasets; designed and optimized a neural network-based recommendation system, achieving **40% higher accuracy** over alternatives.

Full-Stack & AI Developer — Freelance

Oct 2020 – Jul 2021

- Led a **3-member team** across **25+ international deliveries** spanning AI/ML, cloud, and full-stack builds.

TECHNICAL SKILLS

Languages & Data: Python, SQL, Bash, PostgreSQL, MySQL, MongoDB, Redis

LLM, Agents & RAG: LangGraph, Langflow, LangChain, CrewAI, MCP, LiteLLM, OpenAI SDK, Gemini API, Hugging Face Transformers, function calling, prompt engineering, structured outputs (Pydantic), FAISS, Pinecone, Qdrant, pgvector, hybrid search & reranking, Ragas / LLM-as-judge evaluation, Langfuse (prompt registry, tracing)

AI/ML & Computer Vision: PyTorch (DDP), TensorFlow, Diffusion Models, GANs, Transformers/ViTs, Whisper, Multimodal AI, Model Fine-Tuning & Evaluation, OpenCV, FFmpeg, NumPy, Pandas

Backend & Infrastructure: FastAPI, Django, Flask, REST, JWT/RBAC, Docker, AWS, Nginx, Linux, CUDA, GitLab CI/CD, CloudFront, MLflow, DVC

Engineering Practice: SOLID principles, clean architecture, Ruff, uv, pytest, type hints, asyncio, code reviews, Git branching workflows

PROJECTS, EDUCATION & ACHIEVEMENTS

Projects (85+ on GitHub): Conversational AI Notetaker — live multi-person conversation assistant that takes the topic as context, extracts keywords as people speak, and surfaces notes in the UI in real time · AgenticJobHunt — multi-agent resume analysis and job discovery (LangGraph, deepagents, LiteLLM, Qdrant) · KDC-MLflow-DVC — experiment tracking and data versioning for reproducible ML runs

Education: **M.Tech, AI & Robotics** and **B.Tech, Computer Science & Engineering** — Gautam Buddha University, Greater Noida — GPA 8.0/10 (both)

Publication & Achievements: Improved Two-Stage Generative Adversarial Networks for Example Attack — peer-reviewed, 2023 ([DOI](#)) · GAN Specialization (DeepLearning.AI) · 15+ certifications (Google, IBM, LinkedIn) · AI/ML Instructor, ROBO-AI